PROBLEM 46

Bumpy Taxiway

Original statement · Problems.pdf, p. 31

Resonance & Damping

An airplane is taxiing over a bumpy runway of vertical profile $y_1 = \sin(x)$ with a horizontal velocity c. The vertical motion of the airplane can be modeled as that of a heavy particle P of mass m supported on a landing gear of nominal length L when unloaded. Consider that the landing gear has stiffness constant k and negligible damping.

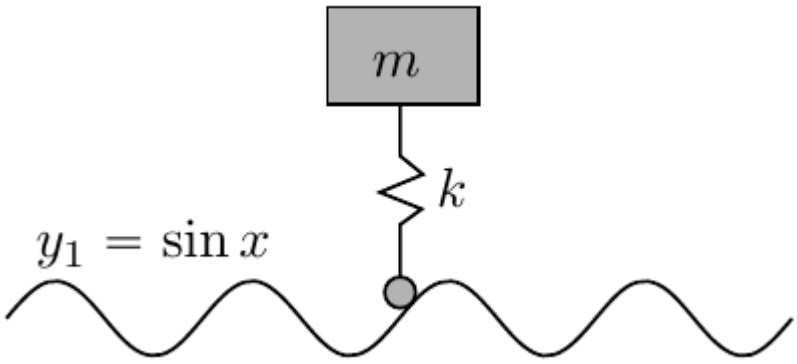


Figure: Mass m on a spring k whose lower end follows the bumpy runway profile y1 = sin x (Problems.pdf, p. 31).

(a) At what taxiing speed is the system in resonance with the bumpy road?

(b) Find the expression of the amplitude of the airplane oscillations when the taxiing velocity c is very high. Hint: the expression depends on c.

(c) Consider now that the landing gear has damping constant λ. How does the results above change? Consider the underdamped, critical, and overdamped cases. What value of λ would you choose for this application and why? If the mass of the aircraft changes (e.g. more or less payload), does this affect your conclusion?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** the fuselage is a particle of mass m that moves only vertically. The landing gear is a spring of stiffness k and unloaded length L whose lower end (the wheel) follows the runway profile $y_1 = \sin x$ at the horizontal speed c.
- **Hypotheses:** the wheel never leaves the ground; the runway profile is read with $x = ct$, so the ground height under the wheel is $y_b(t) = \sin(ct)$ (the profile is written with unit wavenumber, $1\ \mathrm{m}^{-1}$, so the excitation angular frequency is $\Omega = c$ in rad/s when c is in m/s); gravity acts on m; the gear is linear.
- **Parameters and units:** m [kg], k [N/m], L [m], c [m/s], g [m/s^2]; $\omega_0 = \sqrt{k/m}$ [rad/s]; in part (c) a damper of coefficient λ [N s/m] acts between the fuselage and the wheel.
- **Degrees of freedom:** one, the height y of the particle.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ with Oy vertical upward and $\mathbf{r}_P = y\,\mathbf{j}$ (the horizontal motion at speed c does not enter the vertical dynamics).
- **Spring length:** $\ell_s = y - y_b(t)$ with $y_b = \sin ct$; the spring force on P is $-k(\ell_s - L)\,\mathbf{j}$ (upward when compressed).
- **Static equilibrium over flat ground:** $y_s = L - mg/k$; the shifted coordinate $z = y - y_s$ measures the motion relative to it.

PHASE 3

Step-by-Step Mathematical Deduction

Equation of motion

Pedagogical Justification: Newton's second law along Oy with the weight and the elastic force. The constant weight only shifts the equilibrium, so it disappears in the coordinate z.

$$m\ddot{y} = -k\,[y - \sin(ct) - L] - mg$$

Substitute $y = z + L - mg/k$, so that $\ddot{y} = \ddot{z}$ and $-k[y - L] - mg = -kz$:

$$m\ddot{z} = -kz + k\sin(ct) \implies \ddot{z} + \omega_0^2\, z = \omega_0^2 \sin(ct)$$

(a) Resonance

> **Pedagogical Justification:** This is a harmonic oscillator forced at the angular frequency $\Omega = c$ by the base motion (Notes Sec. 5.4). Resonance occurs when the forcing frequency equals the natural frequency.

$$c_{res} = \omega_0 = \sqrt{\frac{k}{m}}$$

With the profile wavelength 2π (unit wavenumber), the resonant speed is $c = \sqrt{k/m}$; for a general wavelength Λ it would be $c = \omega_0 \Lambda / 2\pi$.

(b) Amplitude at very high speed

Try the particular solution $z_p = A\sin(ct)$, for $c \neq \omega_0$:

$$\ddot{z}_p = -c^2 A \sin(ct) \implies (\omega_0^2 - c^2)A = \omega_0^2 \implies A = \frac{\omega_0^2}{\omega_0^2 - c^2}$$

The amplitude of the oscillation of the airplane (the homogeneous part, at ω_0, is the transient set by the initial conditions) is

$$|A(c)| = \frac{\omega_0^2}{|\omega_0^2 - c^2|} \xrightarrow[c \gg \omega_0]{} \frac{\omega_0^2}{c^2} = \frac{k}{m\,c^2}$$

For very high speed the amplitude decays as $1/c^2$: the mass cannot follow the fast bumps, and z_p is in antiphase with the road ($A < 0$ for $c > \omega_0$).

(c) Landing gear with damping λ

> **Pedagogical Justification:** The damper acts on the relative velocity between the fuselage and the wheel, $-\lambda(\dot{y} - \dot{y}_b)$. Define $a = \lambda/m = 2\zeta\omega_0$ with the damping ratio $\zeta = \lambda/(2\sqrt{km})$.

$$\ddot{z} + a\,\dot{z} + \omega_0^2 z = \omega_0^2 \sin(ct) + a\,c\cos(ct)$$

Homogeneous solution (transient). The characteristic equation $r^2 + a\,r + \omega_0^2 = 0$ has roots $r = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$:

- Underdamped, $\lambda < 2\sqrt{km}$ ($\zeta < 1$): $z_h = e^{-\zeta\omega_0 t}\,(C_1\cos\omega_d t + C_2\sin\omega_d t)$ with $\omega_d = \omega_0\sqrt{1-\zeta^2}$.
- Critical, $\lambda = 2\sqrt{km}$ ($\zeta = 1$): $z_h = (C_1 + C_2 t)\,e^{-\omega_0 t}$, the fastest return without oscillation.
- Overdamped, $\lambda > 2\sqrt{km}$ ($\zeta > 1$): $z_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$ with $r_{1,2} < 0$ real, slower return.

Steady state. Write the forcing as the imaginary part of $(\omega_0^2 + iac)\,e^{ict}$ and $z_p = \mathrm{Im}[H\,e^{ict}]$:

$$H = \frac{\omega_0^2 + i\,ac}{\omega_0^2 - c^2 + i\,ac}, \qquad |H|^2 = \frac{\omega_0^4 + a^2c^2}{(\omega_0^2 - c^2)^2 + a^2c^2}$$

The steady amplitude is $|H|$, and it is always finite. At resonance, $c = \omega_0$:

$$|H(\omega_0)| = \sqrt{1 + \frac{\omega_0^2}{a^2}} = \sqrt{1 + \frac{1}{4\zeta^2}}$$

At very high speed $|H| \approx a\,c/c^2 = a/c = \lambda/(mc)$: it decays only as $1/c$, slower than the undamped $1/c^2$ of part (b).

Choice of λ and effect of the payload

- Damping removes the resonant divergence ($|H(\omega_0)|$ finite) and kills the transient quickly, but the price is the slower attenuation $1/c$ of the bumps at high speed. A compromise is ζ below 1 (not overdamped): $\zeta = 0.5$ gives a resonant peak of $\sqrt{2} = 1.41$ and $\zeta = 0.7$ gives

1.22, whereas critical damping slows the response and transmits more at high speed.

- Changing the mass changes both $\omega_0 = \sqrt{k/m}$, so the resonant speed $c_{res} = \sqrt{k/m}$ is lower for a heavier aircraft, and $\zeta = \lambda/(2\sqrt{km}) \propto m^{-1/2}$, so a heavier aircraft is less damped. Since λ is fixed hardware, it must be chosen so that ζ stays in the useful range (for example $\zeta \gtrsim 0.3$) for the heaviest payload; the lightest payload then gets a larger ζ and a higher resonant speed.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional check:** $[\omega_0^2] = [k/m] = \mathrm{s}^{-2} = [c^2]$ with c in rad/s; $[a] = [\lambda/m] = \mathrm{N\,s\,m^{-1}/kg} = \mathrm{s}^{-1}$, so $\zeta = a/(2\omega_0)$ is dimensionless.
- **Limits:** $c \to 0$ gives $|A| \to 1$, the airplane follows the road quasi-statically; $\lambda \to 0$ in $|H|$ recovers $|A|$ of part (b); $\zeta \to \infty$ gives $|H| \to 1$ for slow runways and $\to \lambda/(mc)$ at high speed.
- **Symbolic check (sympy):** the undamped particular solution and the damped steady state $z_p = \mathrm{Im}[He^{ict}]$ satisfy their equations with zero residual, and $|H(\omega_0)|^2 = 1 + \omega_0^2/a^2$.

Source: Problems.pdf (Page 31)